

## Chapter 2 - Modules

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### Exercise 1

Since  $\gcd(m, n) = 1, \exists u, v \in \mathbb{Z}, \text{ s.t.}$

$$mu + nv = 1$$

Now for  $\forall x \otimes y \in (\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}),$

$$\begin{aligned} x \otimes y &= (mu + nv)(x \otimes y) \\ &= mu(x \otimes y) + nv(x \otimes y) \\ &= 0 \end{aligned}$$

□

### Exercise 2

Consider the exact sequence

$$\mathfrak{a} \xrightarrow{f} A \xrightarrow{g} A/\mathfrak{a} \rightarrow 0$$

By *Proposition 2.18*, we have the following exact sequence

$$\mathfrak{a} \otimes M \xrightarrow{f \otimes 1} A \otimes M \xrightarrow{g \otimes 1} (A/\mathfrak{a}) \otimes M \longrightarrow 0$$

Notice there exist trivial isomorphism  $\mathfrak{a} \otimes M \cong \mathfrak{a}M, A \otimes M \cong M$ . So we can have the exact sequence

$$\mathfrak{a}M \xrightarrow{\varphi} M \xrightarrow{\psi} (A/\mathfrak{a}) \otimes M \rightarrow 0$$

With  $\text{Im } \varphi = \text{Ker } \psi, \text{Im } \psi = (A/\mathfrak{a}) \otimes M$ . Which is to say

$$\text{Im } \psi = (A/\mathfrak{a}) \otimes M, \text{Ker } \psi = \mathfrak{a}M$$

So by *Module Isomorphism Theorem*,

$$(A/\mathfrak{a}) \otimes M \cong M/\mathfrak{a}M$$

□

### Exercise 3

We follow from the hint: For any finitely generated  $A$ -module  $M_0$ , let  $\mathfrak{m}$  be the maximal ideal of  $A$  and  $k = M_0/\mathfrak{m}$  be the field derived from quotient. Now by *Ex-2.2*,  $k \otimes_A M_0 \cong M_0/\mathfrak{m}M_0$ . So by *Nakayama Lemma*,  $k \otimes_A M_0 = 0$  will indicate  $M_0 = 0$ .

Back to the problem:

$$\begin{aligned} 0 &= k \otimes_A k \otimes_A \otimes_A M \otimes_A N \\ &\cong k \otimes_A (k \otimes_A M) \otimes_A N \\ &\cong k \otimes_A M_k \otimes_A N \\ &\cong M_k \otimes_A (k \otimes_A N) \\ &\cong M_k \otimes_A N_k \end{aligned}$$

**Exercise 4**

For any  $A$ -module  $N$ , consider

$$f : N \longrightarrow N'$$

which is injective. Now

$M$  is flat

$\Leftrightarrow$

$$f \otimes 1 : N \otimes M \longrightarrow N' \otimes M \text{ is injective}$$

$\Leftrightarrow$  (Be considering their kernel)

$$\forall i \in I, f \otimes 1 : N \otimes M_i \longrightarrow N' \otimes M_i \text{ is injective}$$

□

**Exercise 5**

We can view  $A, xA, x^2A, \dots$  as a sequence of  $A$ -module, and  $A[x]$  is their direct sum where multiplication in the  $A$ -algebra structure defined in the trivial way.

**Exercise 6**

Define

$$\varphi : A[x] \otimes_A M \longrightarrow M[x]$$

by mapping

$$p(x) \otimes m \longmapsto p(x) \cdot m$$

Obviously  $\text{Ker } \varphi = 0$ ,  $\text{Im } \varphi = M[x]$ , so  $\varphi$  is an isomorphism.

**Exercise 7**

- (i) If  $a(x), b(x) \in A[x], a(x)b(x) \in \mathfrak{p}[x]$ . Let  $a(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0, b(x) = b_n x^n + b_{n-1} x^{n-1} + \dots + b_1 x + b_0$ , then

$$(a_n x^n + \dots + a_1 x + a_0)(b_n x^n + \dots + b_1 x + b_0) \in \mathfrak{p}[x]$$

$$a_n b_n \in \mathfrak{p} \Rightarrow a_n \in \mathfrak{p} \text{ or } b_n \in \mathfrak{p}$$

Suppose  $a_n \in \mathfrak{p}$ , then let  $a'(x) = a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0$  and we have

$$a'(x)b(x) \in \mathfrak{p}[x]$$

Repeat the process above and eventually

$$a(x) \in \mathfrak{p}[x] \text{ or } b(x) \in \mathfrak{p}[x]$$

□

- (ii) Not necessarily.

Let  $A = k$  be a field and  $\mathfrak{m} = (0)$ . Obviously  $\mathfrak{m}[x] = (0)$  is not a maximal ideal in  $k[x]$ .

**Exercise 8**

- (i) Consider the injective  $A$ -module homomorphism

$$\varphi : R \longrightarrow R'$$

Since  $M$  is flat,

$$\varphi \otimes 1 : R \otimes_A M \longrightarrow R' \otimes_A M$$

is injective, therefore

$$\varphi \otimes 1 \otimes 1 : R \otimes_A M \otimes_A N \longrightarrow R' \otimes_A M \otimes_A N$$

is injective, i.e.

$$\varphi \otimes 1 : R \otimes_A (M \otimes_A N) \longrightarrow R' \otimes_A (M \otimes_A N)$$

is injective. So  $M \otimes_A N$  is flat.

(ii) Consider a injective  $B$ -module homomorphism

$$\varphi : M \longrightarrow M'$$

Since  $N$  is a flat  $B$ -module homomorphism,

$$\varphi \otimes_B 1 : M \otimes_B N \longrightarrow M' \otimes_B N$$

is injective. To prove  $N$  is a flat  $A$ -module, we need to show

$$\varphi \otimes_A 1 : M \otimes_A N \longrightarrow M' \otimes_A N$$

is injective, i.e.  $\text{Ker } \varphi \otimes_A 1 = \{0\}$ .

Notice  $\text{Ker } \varphi \otimes_A 1 \subseteq \text{Ker } \varphi \otimes_B 1 = \{0\}$ , we're done.  $\square$

### Exercise 9

Denote

$$0 \rightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$$

We know

$$M \cong \text{Coim } g \oplus \text{Ker } g \cong \text{Im } g \oplus \text{Ker } g$$

Be the exactness,  $\text{Ker } g = \text{Im } f$ ,  $\text{Im } g = M''$ , which are both finitely generated. So  $M$  is finitely generated.  $\square$

### Exercise 10

Since  $M/\mathfrak{a}M \longrightarrow N/\mathfrak{a}N$  is surjective,

$$M \longrightarrow M/\mathfrak{a}M \longrightarrow N/\mathfrak{a}N$$

is surjective. Therefore  $N = \mathfrak{a}N + u(M)$ , and by *Corollary 2.7*,  $u(M) = N$ .  $\square$

### Exercise 11

(The ring multiplication here is Cartesian product!!!)

(i) When  $A^m \cong A^n$  : We use the method in the textbook.

Let  $\mathfrak{m}$  be a maximal ideal of  $A$ , and denote the isomorphism by

$$\phi : A^m \longrightarrow A^n$$

By checking the kernel and image, we can easily see that

$$1 \otimes_A \phi : (A/\mathfrak{m}) \otimes_A A^m \longrightarrow (A/\mathfrak{m}) \otimes_A A^n$$

is an isomorphism by *Proposition 2.18*. By *Exercise 2.2* we know  $(A/\mathfrak{m})^m \cong A^m/\mathfrak{m} \cong (A/\mathfrak{m}) \otimes A^m$ ,  $(A/\mathfrak{m})^n \cong A^n/\mathfrak{m} \cong (A/\mathfrak{m}) \otimes A^n$  are two isomorphic linear space over the field  $k = A/\mathfrak{m}$ . Hence

$$\text{Dim } k^m = \text{Dim } k^n, m = n$$

(ii) When  $\phi : A^m \longrightarrow A^n$  is surjective: Similar to (i), we obtain a surjective  $k$ -linear space homomorphism by using *Proposition 2.18*

$$k^m \longrightarrow k^n$$

So  $m > n$  is obtained by the property of linear space homomorphisms.

(iii) When  $\phi : A^m \longrightarrow A^n$  is injective: Yes.

Note that we're no longer able to use *Proposition 2.18* to convert ring homomorphisms into linear space homomorphisms.

Suppose  $m > n$ , we can define  $A^n$  to be the submodule of  $A^m$  by taking the first  $n$  terms and extend the range of  $\phi$  to  $A^m$ . Now

$$\phi(A^m) \subseteq A^n \subseteq A^m = \mathfrak{a}A^m$$

where  $\mathfrak{a} = A$  is viewed as an ideal.

Since  $A^m$  is a finitely generated  $A$ -module, by *Proposition 2.4*,  $\exists a_1, a_2, \dots, a_s \in A$ , s.t.

$$\phi^s + a_1\phi^{s-1} + \dots + a_s = 0, a_s \neq 0$$

Specifically,

$$(\phi^s + a_1\phi^{s-1} + \dots + a_{s-1}\phi + a_s\text{Id})(0, \dots, 0, 1) = 0$$

Notice  $\forall a \in A^m$ , because  $m > n$ , the last term of  $\phi(a)$  must be 0. So

$$(\phi^s + \dots + a_{s-1}\phi)(0, \dots, 0, 1) = (\dots, 0)$$

But

$$a_s\text{Id}(0, \dots, 0, 1) = (\dots, a_s)$$

Hence, it's impossible to have  $(\phi^s + a_1\phi^{s-1} + \dots + a_{s-1}\phi + a_s\text{Id})(0, \dots, 0, 1) = 0$  So  $m \leq n$ .

□

### Exercise 12

Let  $A^n = \langle e_1, e_2, \dots, e_n \rangle$  where

$$\begin{cases} e_1 = (1, 0, 0, \dots, 0) \\ e_2 = (0, 1, 0, \dots, 0) \\ \vdots \\ e_n = (0, 0, 0, \dots, 1) \end{cases}$$

and choose  $u_k \in M$ , s.t.  $\phi(u_k) = e_k$ . We can prove every element in  $M$  can be expressed in the sum of element in  $\langle u_1, u_2, \dots, u_n \rangle, \text{Ker } \phi$ . Also,  $\text{Ker } \phi \cap \langle u_1, \dots, u_n \rangle = \{0\}$ , so  $M = \text{Ker } \phi \oplus \langle u_1, \dots, u_n \rangle$ .

### Exercise 13

(i)  $\text{Ker } g = 0$ , so  $g$  is injective.

(ii) Define

$$p : N_B \longrightarrow N$$

by

$$b \otimes y \longmapsto by$$

and it's trivial that

$$N_B = \text{Im } g \oplus \text{Ker } p$$

□

### Exercise 14

To prove  $\mu_i = \mu_j \circ \mu_{ij}$ , we only need to understand that in the structure of  $M = C/D$ , elements are the equivalence classes in  $C$  where equivalence relation defined by homomorphisms. The action of  $\mu_{ij}$  is simply sending one element to another in the same equivalence class.

After one understand this, the problem becomes rather simple.

### Exercise 15

(i) This is to say that  $\mu : C \longrightarrow C/D$  is surjective, which is obviously correct.

(ii) This is automatically obtained by the definition of  $D$ .

### Exercise 16

Because any such homomorphism that satisfy

$$\forall i \in I, \alpha_i = \alpha \circ \mu_i$$

is unique determined by the inner structure of  $M_i, N, M$ .

**Exercise 17**

Because in the notation of *Exercise 2.14*,  $D = \{0\}$  and therefore

$$\lim_{\rightarrow} M_i = C/D = C = \sum_{i \in I} M_i$$

□

**Exercise 18**

$$\begin{array}{ccc} M_i & \xrightarrow{\phi_i} & N_i \\ \downarrow \mu_i & & \downarrow \nu_i \\ M & \xrightarrow{\phi} & N \end{array}$$

Because from those  $\nu_i \circ \phi_i$  defining the direct limit homomorphism  $M_i \rightarrow N$ , the unique homomorphism  $\phi$  is obtained by the universal property of direct limit.

**Exercise 19**

$$\begin{array}{ccccc} M_i & \xrightarrow{\phi_i} & N_i & \xrightarrow{\psi_i} & P_i \\ \downarrow \mu_i & & \downarrow \nu_i & & \downarrow \xi_i \\ M & \xrightarrow{\phi} & N & \xrightarrow{\psi} & P \end{array}$$

By *Exercise 2.15*, each elements in  $M$  can be expressed in the form  $\mu_i(x_i)$ , and  $\text{Ker } \psi = \{\nu_i \text{Ker } \psi_i\}_{i \in I}$ . Therefore

$$\begin{aligned} \text{Im } \phi &= \{\text{Im } \phi \circ \mu_i\}_{i \in I} \\ &= \{\text{Im } \nu_i \circ \phi_i\}_{i \in I} \\ &= \{\nu_i \text{Im } \phi_i\}_{i \in I} \\ &= \{\nu_i \text{Ker } \psi_i\}_{i \in I} \\ &= \text{Ker } \psi \end{aligned}$$

□

**Exercise 20**

(FAILED)

$$\begin{array}{ccc} M_i \times N & \xrightarrow{\quad} & M_i \otimes N \\ \downarrow & & \downarrow \\ (\lim_{\rightarrow} M_i) \times N & \xrightarrow{g} & \lim_{\rightarrow} (M_i \otimes N) \\ & \searrow & \uparrow \psi \quad \downarrow \phi \\ & & (\lim_{\rightarrow} M_i) \otimes N \end{array}$$

We follow from the hint: The proof of  $g$  being  $A$ -bilinear is trivial.

(I don't understand exactly how to prove  $\psi$  and  $\phi$  is inverse to each other, more precisely to check how one element was sent to under this homomorphism.)

**Exercise 21**

(i) (Additive structure)

$A = \lim_{\rightarrow} A_i$  naturally inherit the structure of  $\mathbb{Z}$ -module by the definition of direct limit.

(ii) (*Close under multiplication*)

For  $\forall \alpha, \beta \in A$ , by *Exercise 2.15* there exist  $\mu_i \in \text{Hom}(A_i, A), \mu_j \in \text{Hom}(A_j, A), x_i \in A_i, x_j \in A_j$  such that

$$\alpha = \mu_i(x_i), \beta = \mu_j(x_j)$$

Since  $I$  is a directed set,  $\exists k \in I$ , s.t.  $i \leq k, j \leq k$ . Therefore we can have

$$\begin{aligned} \alpha\beta &= \mu_i(x_i)\mu_j(x_j) \\ &= \mu(x_i)\mu(x_j) \\ &= \mu(\mu_{ik}(x_i))\mu(\mu_{jk}(x_j)) \\ &= \mu_k(\mu_{ik}(x_i))\mu_k(\mu_{jk}(x_j)) \end{aligned}$$

and define

$$\alpha\beta = \mu_k(\mu_{ik}(x_i)\mu_{jk}(x_j))$$

Such definition of multiplicative result is independent of the choice of  $k$  as well as the representation of  $x_i, x_j$ , by property of directed limit. Therefore it's well-defined.

(iii) (*Multiplicative identity*)

The identity for  $A$  is  $\mu(1)$ , where  $1$  is the identity in an arbitrary  $A_i$  (since their images are equal).

(iv) (*Homomorphism*)

The fact that  $\mu : \bigotimes A_i \longrightarrow A$  being a ring homomorphism is naturally obtained from the definition in (ii).

□

## Exercise 22

$$\begin{array}{ccc} A_i & \xrightarrow{\quad\quad\quad} & \text{Nil}(A_i) \\ \downarrow & & \downarrow \\ \lim_{\rightarrow} A_i & \xrightarrow{\quad\quad\quad} & \lim_{\rightarrow} (\text{Nil}(A_i)) \\ & \nearrow & \\ & \text{Nil}(\lim_{\rightarrow} A_i) & \end{array}$$

(i)  $\text{Nil}(\lim_{\rightarrow} A_i) \subseteq \lim_{\rightarrow} (\text{Nil}(A_i))$ :

$\forall x \in \lim_{\rightarrow} A_i, \exists x_i \in A_i, \mu_i \in \text{Hom}(A_i, A)$ , s.t.  $\mu_i(x_i) = x$ . Now suppose  $x \in \text{Nil}(\lim_{\rightarrow} A_i)$ , then  $\exists k \in \mathbb{Z}$ , s.t.  $x^k = 0$ . By the homomorphism structure defined in *Exercise 2.21*, that is to say

$$x_i^k = 0, x_i \in \text{Nil}(A_i)$$

Therefore  $\text{Nil}(\lim_{\rightarrow} A_i) \subseteq \lim_{\rightarrow} (\text{Nil}(A_i))$ .

(ii)  $\lim_{\rightarrow} (\text{Nil}(A_i)) \subseteq \text{Nil}(\lim_{\rightarrow} A_i)$ :

The image of a nilpotent in the direct limit preserves to be nilpotent, that is to say

$$\lim_{\rightarrow} (\text{Nil}(A_i)) \subseteq \text{Nil}(\lim_{\rightarrow} A_i)$$

□

## Exercise 23

$$\begin{array}{ccc} \prod_{\lambda \in \Lambda} B_{\lambda} & \xrightarrow{\quad\quad\quad} & P \\ & \searrow \quad \swarrow & \\ & \bigotimes_{\lambda \in \Lambda} B_{\lambda} = \lim_{\substack{\rightarrow \\ J \subseteq \Lambda}} B_J & \end{array}$$

The map

$$\prod_{\lambda \in \Lambda} B_{\lambda} \longrightarrow \varinjlim_{J \subseteq \Lambda} B_J \longrightarrow P$$

gives a  $\{B_{\lambda}\}_{\lambda \in \Lambda}$ -multilinear map, therefore induce a surjective homomorphism

$$\Phi : \bigotimes_{\lambda \in \Lambda} B_{\lambda} \longrightarrow \varinjlim_{J \subseteq \Lambda} B_J$$

We can easily check that  $\text{Ker } \Phi = 0$ , so  $\Phi$  is injective. So  $\Phi$  is an isomorphism, and  $\bigotimes_{\lambda \in \Lambda} B_{\lambda} \cong \varinjlim_{J \subseteq \Lambda} B_J$  is the same up to isomorphism.  $\square$

**I'll skip the rest of this chapter since I haven't learnt anything about tor functor and homological algebra :(**